National University of Computer and Emerging Sciences



Lab Manual 07
Artificial Intelligence

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Objectives:

- ✓ Hill Climbing Algorithm for optimization problems

Activities:

Activity 1 (Searching Algorithms):

Hill Climbing Algorithm

Local search algorithms are a class of search techniques that focus on finding an optimal solution by iteratively improving a single candidate solution. Unlike traditional search algorithms that explore a tree or graph and maintain a record of all explored paths, local search algorithms maintain only the current state in memory. These algorithms are particularly useful for optimization problems where the goal is to find the best possible state rather than the exact path to that state.

Hill Climbing is one of the simplest form of local search algorithm that attempts to improve the current solution by moving to a better neighbouring state. It is widely used for problems with large state spaces.

Hill Climbing evaluates only the neighbouring states of the current position and moves to the one that appears to be the best. If no neighbour improves the current state, the algorithm stops, potentially at a local maximum rather than the global optimum.

It is analogous to: *A person climbing a hill in fog, always moving upward.*

Important Concepts

State

A candidate solution.
Example: Position on a hill.

Neighbour

A state reachable from the current state.
Example: Nearby positions on the hill.

Evaluation Function

Measures how good a state is.

$$f\ (state)$$

$$Higher\ value\ \rightarrow Better\ State$$

Hill Climbing Algorithm Steps

1. Choose an initial state
2. Repeat:
   - Evaluate current state
   - Generate neighbours
   - Choose best neighbour
3. If best neighbour is better:
   - Move to neighbour
4. Otherwise:

- Stop (local optimum)

Variants of Hill Climbing

Simple Hill Climbing

Examines neighbours one by one.

Steepest-Ascent Hill Climbing

Evaluates all neighbours and chooses the best.

Stochastic Hill Climbing

Chooses random improving neighbour.

Limitations of Hill Climbing

Hill climbing may get stuck in:

Local Maximum

Better than neighbours but not global best.

Plateau

Flat area with equal values.

Ridge

Narrow path difficult for greedy moves.

Techniques to Improve Hill Climbing

- Random Restart Hill Climbing
- Sideways Moves
- Stochastic Hill Climbing

Activity 2 (Problem Statements / Tasks):

1. *Hill Climbing for Maximum Value*

Consider the following state landscape:

```
landscape = {
    1: [2, 3],
    2: [4, 5],
    3: [5],
    4: [7],
    5: [6],
    6: [8],
    7: [],
    8: []
}
```

Requirements

- Implement the Hill Climbing algorithm.
- Start the search from state 1.
- At each step:
  - Evaluate neighbours
  - Move to the best neighbour.

Display

- Visited states during the search
- Current state at each step
- Final local maximum reached

Explain:

- Why the algorithm stops at the final state.

2. *Hill Climbing with Random Restart*

Hill Climbing can sometimes get stuck in local maxima.
Modify the Hill Climbing algorithm to implement Random Restart Hill Climbing.

Requirements

- Run Hill Climbing multiple times with different starting states.
- Randomly select starting states from the landscape.
- Run the algorithm 5 times.

Display

- Starting state for each run
- Local maximum obtained in each run
- Best solution among all runs
- Total number of restarts

Explain

- Why Random Restart can produce better solutions than a single run.

3. *Hill Climbing for Function Optimization*

Consider the mathematical function:

$$f(x) = -(x-3)^2 + 9$$

This function has a maximum value at x = 3.

Requirements

- Implement Hill Climbing to find the maximum value.
- Generate neighbours using:
  x + 1
  x - 1
- Start the search from:
  x = 0

Display

- Current value of x at each step
- Value of f(x)
- Neighbour evaluated
- Final maximum value obtained

Expected Result

The algorithm should converge near:

x = 3
f(x) = 9

4. *Stochastic Hill Climbing for Function Optimization*

- Maximize $f(x) = -(x-4)^2 + 16$, where x ranges from 0 to 8.
- Randomly select improving neighbours.

- Run the algorithm five times using different starting values.
- Compare the paths and identify the best solution.